

#	Week	Tr Unit	Topic Covered	Practical Idea		No	Course Pedagogy	Marks
1	1		Component Architecture, Frontend Architecture	Set up React app, build static UI with reusable components			Internal	
2	1		State Management, Routing	Add state management and React Router for multi-page feel		1	Continuous Evaluation	5
3	1, 2		API Integration, Rendering Lifecycle	Connect React to REST API, handle loading and error states		2	Mid Practical Exam	10
4	2		Performance Optimization, Lazy Loading	Add lazy loading and code splitting, measure with dev tools		3	MOOC Course (Coursera)	10
5	3		RESTful API Design, Middleware	Build Node/Express server with REST endpoints and middleware				25
6	3		MongoDB, Schema Enforcement, Validation	Connect MongoDB, perform CRUD with Mongoose schema			External	
7	3		JWT Auth, Error Handling, Caching	Add JWT auth, input validation, basic in-memory caching		1	Viva	5
8	4		Stateless Services, Connection Pooling	Wire full React + Node + MongoDB app, optimize DB queries		2	University Practical Exam	20
9	4		Load Balancing, Performance Load Testing	Simulate load, measure RPS and latency on the running app				25
10	4		Asynchronous Processing, Message Queues	Add background job (email/notification) using a queue				
11	5		Docker, Containerization	Containerize full app using Docker and Docker Compose				
12	5		CI/CD Pipeline, Deployment Strategies	Set up GitHub Actions – auto test and deploy on push				
13	6		AI Integration, Secure Coding, Responsible AI	Harden app security + integrate one AI API feature				